domain.
(Middle) When the system cools down, transitions to higher-states become unlikely (thinner lines, in the figure, are to show lower probability) and above some threshold, they become so unlikely to be considered as forbidden. Transition to lower states are always allowed.
(Bottom) When we are close to the end of the annealing simulation, the probability to move to higher-energy states is so slim to be negligible: the system can only move to better solutions.

This mimics high-energy systems, where particles (or molecules) move chaotically in all directions.

As the system is cooled down, the distribution changes (figure 17.6.B): going uphill becomes less likely, and some positions above a certain delta (with respect to their cost) become completely unreachable.

Finally, when temperature gets close to the T_{LOW}, the halting temperature (or equivalently, as we'll see, when we are close the maximum number of iterations), going uphill becomes so unlikely to be basically forbidden, and the algorithm can only transition to lower-energy states: it behaves like *hill descent*[9]. Looking at figure 17.6.C, however, you can realize that there is a difference, and transitions to points far away in the domain are allowed, if they correspond to a lower cost. This means the algorithm can still get out of local minima (and it's hopefully even probable that it will!) and converge to a local optimum.